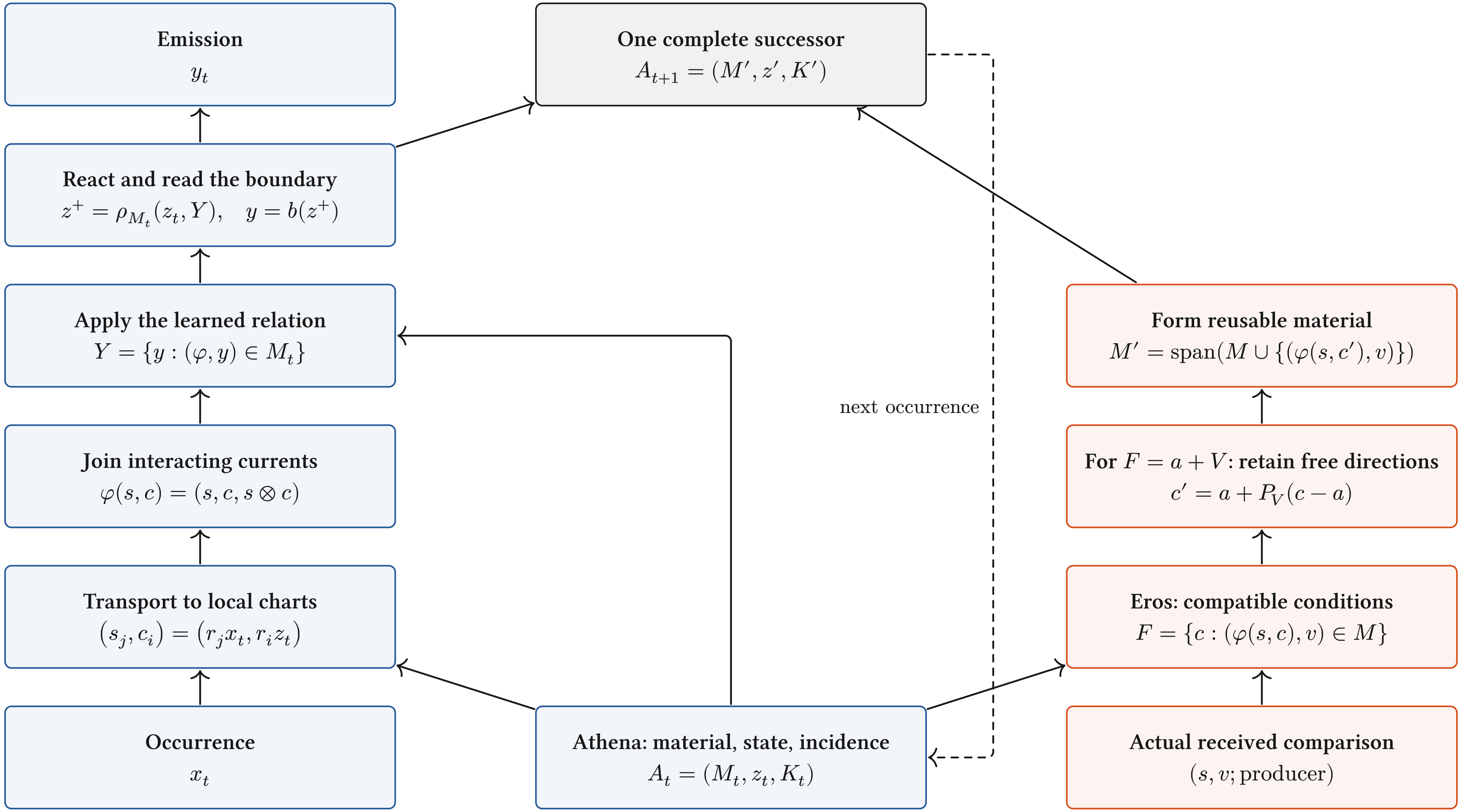
PLATE 01

Athena conducts. Eros develops.

One continuing body; explicit current paths and a joined developmental return.



Definition / source-inspected binding. Blue: conduct. Red: Eros's local relation/condition update. Source, material and the comparison retain their actual producing cut.

Interpretation: full assembly. Local passages compose over admitted incidence. The bilinear relation above is an existing binding; complete contextual assembly remains in progress.

PLATE 02

What a layer does to a space

Finite vector charts: the operation, its categorical role, and the distinctions it retains.

Linear map / “projection”

$$x \mapsto Wx$$

$$V \xrightarrow{W} \operatorname{im} W$$

Linear morphism. Fibres are $x + \ker W$; rank controls the image. It is a projection only when $W^2 = W$ on one space.

Concatenate / retain both

$$(x, y) \mapsto x \oplus y$$

$$\begin{tikzcd} V \arrow[dr, "i_V"] & \\ & V \oplus U \\ U \arrow[ur, "i_U"] & \end{tikzcd}$$

Biproduct injections retain both components. Combining their storage does not yet identify them.

Add / superpose

$$(x, y) \mapsto x + y$$

$$V \oplus V \xrightarrow{\nabla} V$$

The codiagonal identifies $(x + h, y - h)$. Opposed contributions can cancel at this receiver.

Tensor / interact

$$(x, y) \mapsto x \otimes y$$

$$V \times U \xrightarrow{\otimes} V \otimes U$$

Universal bilinear interaction. A bilinear response factors through a linear map out of $V \otimes U$. Gating adds a declared nonlinearity.

Softmax / relative participation

$$p_i = \frac{e^{s_i}}{\sum_j e^{s_j}}$$

$$\mathbb{R}^n/\langle 1 \rangle \xrightarrow{\tilde{\sigma}} \operatorname{int} \Delta^{n-1}$$

Smooth bijection of the score-difference chart with the simplex interior. All $p_i > 0$; common score shifts disappear.

Normalize / choose a section

$$N_0(x) = \sqrt{d}\,\frac{x}{\|x\|}$$

$$\big(\mathbb{R}^d \setminus \{0\}\big)/\mathbb{R}_{>0} \longrightarrow S^{d-1}_{\sqrt{d}}$$

Zero-offset RMS normalization chooses one point per positive ray. Layer centering additionally removes the common-value direction.

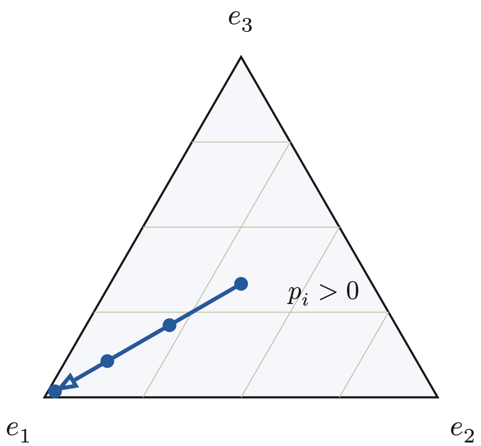
$$N_\varepsilon(x) = \frac{x}{\sqrt{\frac{\|x\|^2}{d} + \varepsilon}}, \quad \varepsilon > 0: \quad \mathbb{R}^d \simeq \left\{ y : \|y\| < \sqrt{d} \right\}$$

Proved-standard, stated finite charts. Positive ε retains radial scale before learned gains: stabilized RMS normalization is not the exact ray quotient above. Zero gain coordinates can erase further information.

Smooth redistribution, selection walls, rank changes

A geometric distinction between moving within a chart and changing the active continuation.

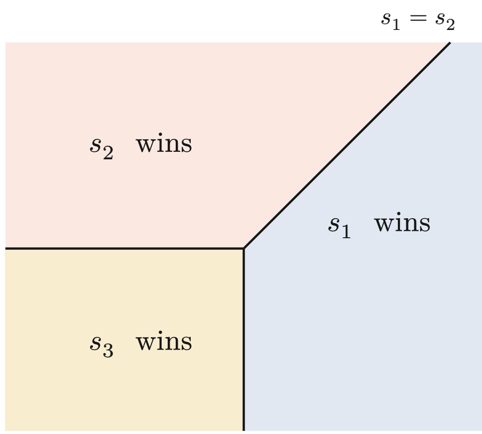
Softmax stays inside the simplex



$s = (\lambda, 0, 0), \quad p = \frac{1}{\exp(\lambda)+2}(\exp(\lambda), 1, 1)$

Every finite λ keeps three positive shares. The blue path approaches a vertex only in the limit.

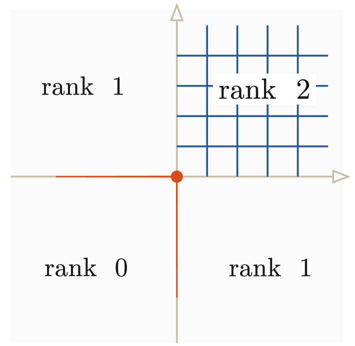
Hard selection crosses a wall



$E(s) = \mathrm{argmax}_i\, s_i, \quad s = (a, b, 0)$

The maximal-score tie walls divide parameter space into regions with different active indices. A tie convention is part of the map.

A nonlinear map changes local rank



$r(x)_i = \max(0, x_i), \quad Dr = \mathrm{diag}\big(1_{x_i>0}\big)$

The four sign chambers form a support lattice. Whole negative directions collapse; the derivative changes at its walls.

$$\begin{tikzcd} & \{1\} \arrow[dr] & \\ \varnothing \arrow[ur] \arrow[dr] & & \{1,2\} \\ & \{2\} \arrow[ur] & \end{tikzcd}$$

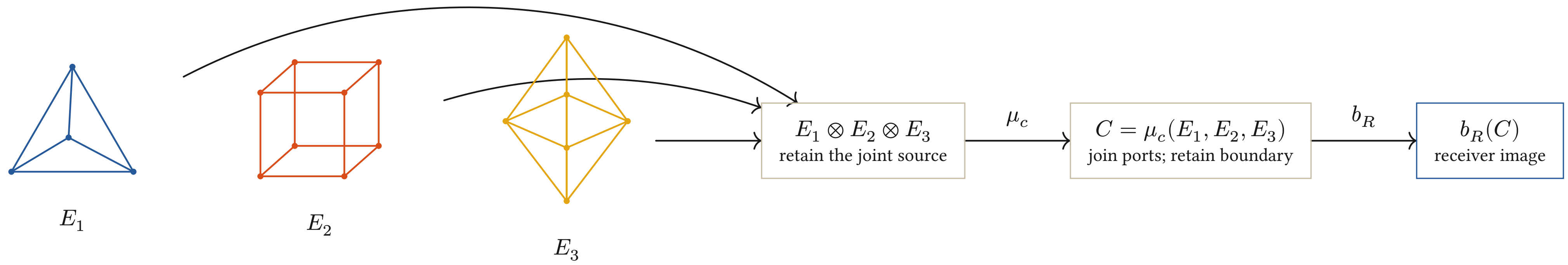
$$M_t = \begin{pmatrix} 1 & 0 \\ 0 & t \end{pmatrix} \longrightarrow \mathrm{rank}\, M_t = \begin{cases} 2 \text{ if } & t \neq 0 \\ 1 \text{ if } & t = 0 \end{cases}$$

Proved-derived finite charts; interpretation of phase. Smooth participation, active-set change and rank loss are different events. Information Chemistry calls a change of the continuing transport class a phase transition; a physical phase claim additionally requires its constitutive law.

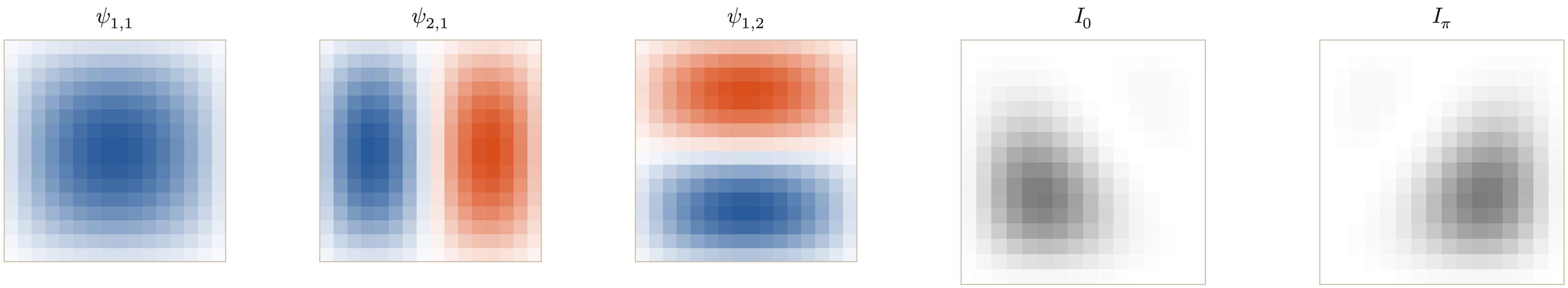
PLATE 04

Information Chemistry: elements into compounds

Typed ports compose; relative phase changes the receiver image of the same constituent modes.



Interpretation. Polyhedra depict ported boundary cells, not semantic labels. The lattice tiles below give a separate explicit realization of constituent composition.



$$b = 4, \quad N = 2^b, \quad (i, j) = (k+1, l+1), \quad k, l \in \{0, \ldots, N-1\}$$

$$\psi_{m,n}(i, j) = \sin\left(\frac{m\pi i}{N+1}\right) \sin\left(\frac{n\pi j}{N+1}\right), \quad N + 1 = 2^4 + 1 = 17$$

$$u_\theta = \psi_{1,1} + \frac{7}{10} e^{i\theta} \psi_{2,1} + \frac{1}{2} \psi_{1,2}, \quad I_\theta = |u_\theta|^2$$

$$|a + b|^2 = |a|^2 + |b|^2 + 2\,\mathrm{Re}\left(a\bar{b}\right)$$

Definition / declared source chart. Four bits address N interior sites per axis; the two Dirichlet boundaries are $N+1$ steps apart. Declared drives $(10, 7, 5)$ relative to the first give $\left(1, \frac{7}{10}, \frac{1}{2}\right)$; the shared intensity bound is $\left(1 + \frac{7}{10} + \frac{1}{2}\right)^2 = \frac{121}{25}$.

Conjecture to develop: one reusable family of typed, parameterized generators can compose every admitted behavioral class. A pattern is a whole diagram modulo its declared future receivers, not a unique solid or a stored output image.

PLATE 05

When diagrams agree, Soulkiller can lift their maps

Compare complete transitions; then restrict at an explicitly declared receiver family.

A commuting architecture

$$
\begin{tikzcd}
A \times X_A \arrow[r, "F_A"] \arrow[d, "q_{\mathrm{in}}"] & Y_A \times T_A \times A \arrow[d, "q_{\mathrm{out}}"] \\
B \times X_B \arrow[r, "F_B"] & Y_B \times T_B \times B
\end{tikzcd}
$$

$q_{\mathrm{out}} F_A = F_B q_{\mathrm{in}}$

Invertible maps: re-expression.
Noninvertible maps: retained fibres.
Unequal paths: explicit defect.

$X \leftarrow W_f \rightarrow Y \leftarrow W_g \rightarrow Z$

$W_{g \circ f} = W_f \times_Y W_g$

Definition / proved-derived composition. Serial joining retains the actual occurrence pullback. The same state map appears before and after each transition.

Soulkiller: realization, then restriction

$$
\begin{tikzcd}
\text{Source diagram } D_S \arrow[r, "i"] & \text{Native realization } D_N \arrow[r, "r"] \arrow[d] & \text{Retained body } D_R \\
& \text{Excite and intervene (actual receiver/history family)} \arrow[ur] \arrow[dl] \arrow[d] \arrow[dr] & \\
\text{Cold witness} & \text{Native material} \arrow[d] & \text{Insufficiency} \\
& \text{Eros} \rightarrow \text{successor Athena} &
\end{tikzcd}
$$

$F_N i = i F_S, \quad r F_N = F_R r$

$\therefore F_R(ri) = (ri) F_S$

Conditional lift. The equations state the required scope. Existing SKE evidence covers its declared families; a full V4.1 binding remains open.

Face-conserving development: $q(U_A(M, v)) = U_B(q(M), v)$

Definition: compatibility law. The two routes preserve the same declared faces when their full difference vanishes. Plates 6 to 9 follow actual string fields through diffusion, images, geometric projection and an exact induced update.

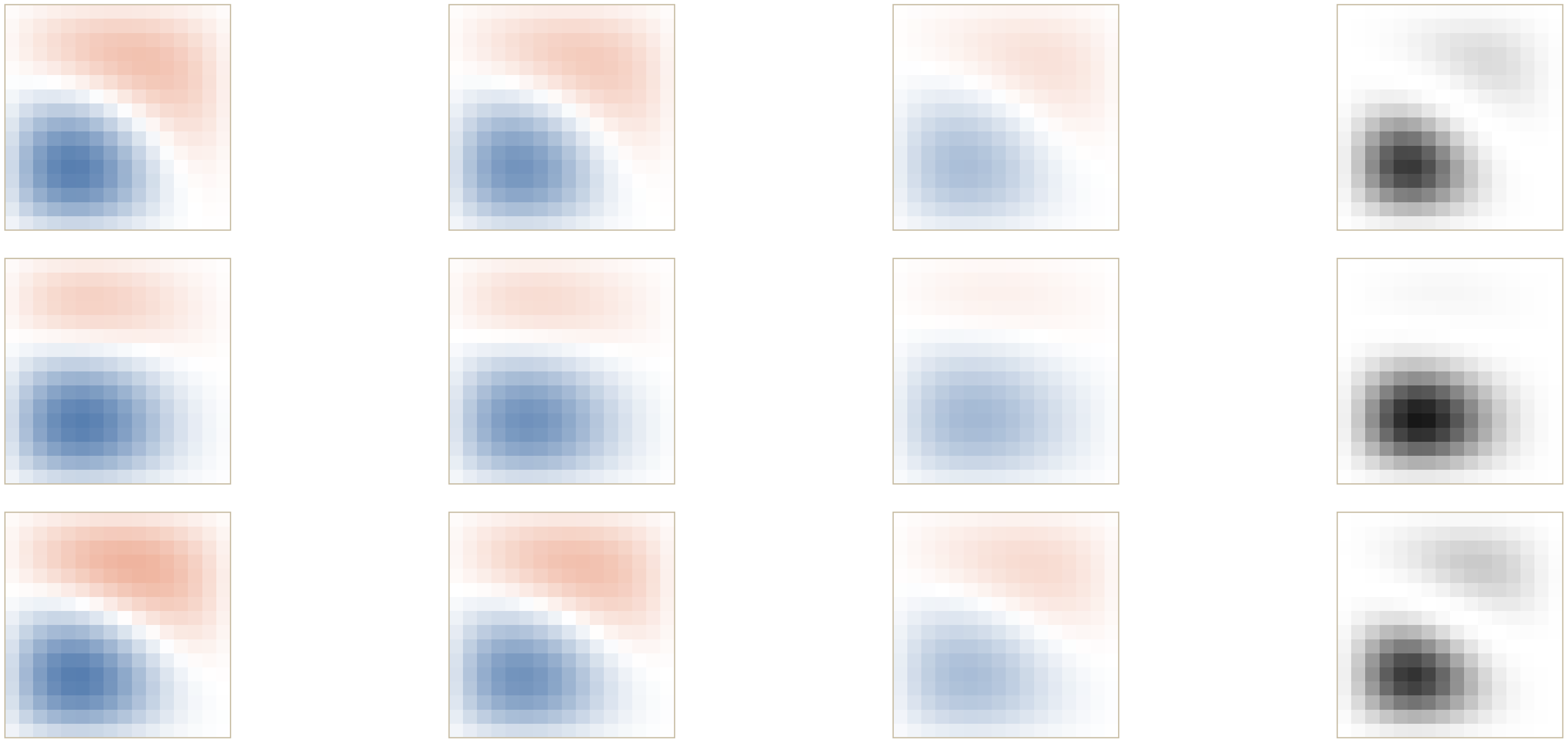
Sources: **Categorical Holonics**; **Elements of Holonics**; the July 19 Information Chemistry record; the August 9 crystal/receiver record; current HNN composition and source owners.
Diagrams: Fletcher 0.5.8, with the repository's geometric notation and CeTZ. Mathematical interpretations and conjecture do not change implementation grades.

PLATE 06

Real strings diffuse into different images

The tetrahedral port weights drive actual fields; each row follows the same diffusion law.

Source inscription	u_0	$u_{\frac{N}{4}}$	u_N	$\frac{I_N}{I_\star}$
1+2=3 chart weight m =7				
2+1=3 chart weight m =7				
1+2=4 chart weight m =8				



$$\lambda = \frac{(a,b,c,1)}{a+b+c+1}, \quad u_0 = \lambda_a\psi_{1,1} + \lambda_b\psi_{2,1} + \lambda_c\psi_{1,2} + \lambda_r\psi_{2,2}$$

$$(Du)_{i,j} = \frac{u_{i-1,j} + u_{i+1,j} + u_{i,j-1} + u_{i,j+1}}{4}, \quad u_k = D^k u_0$$

Definition / explicit lattice realization. $N = 2^4$; exterior samples vanish at the Dirichlet boundary. The factor $\frac{1}{4}$ comes from four neighboring ports. Columns show 0, $\frac{N}{4} = 4$, and $N = 16$ steps. Blue/red: signed current on $[-1,1]$. Gray: $\frac{I_N}{I_\star}$, where $I_\star$ is the maximum over the displayed source/reflection family and all lattice sites; one reference is shared by every intensity tile.

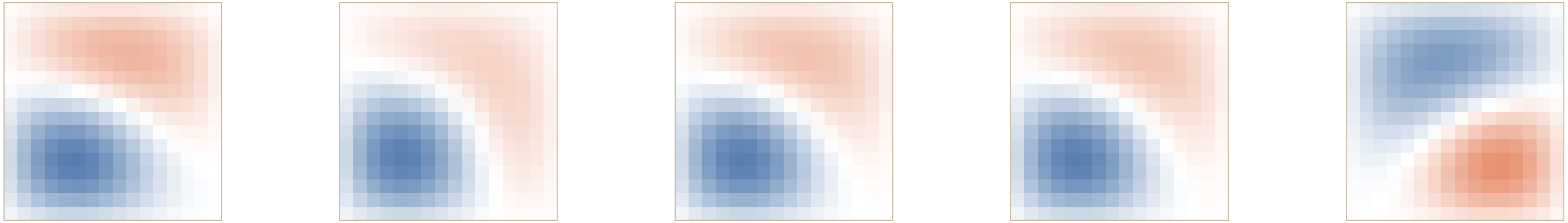
PLATE 07

Reflection and union are visible in the images

Same four modes, same diffusion, different source combinations. The difference is retained as a field.

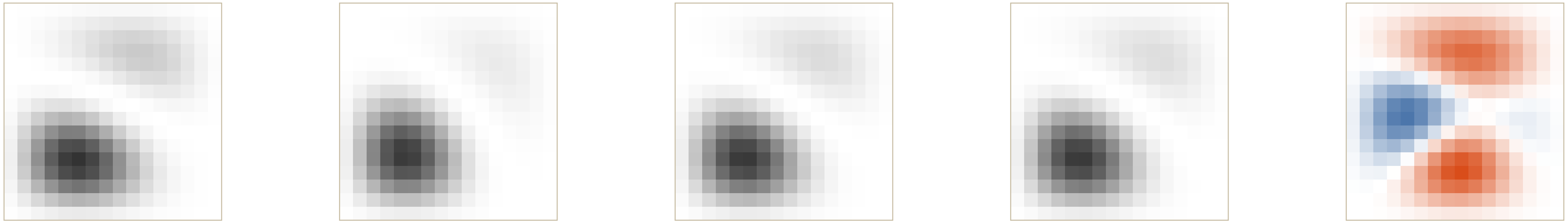
Received	**Reflected**	**Transported union**	**Ordinary chart mean**	**Difference field**
1+2=4	1+2=2	1+2=3	$\left(1, 2, \frac{20}{7}\right)$	ordinary − transported

Current at the initial cut



$42\Delta u_0$

Receiver intensity after N diffusion steps



$\frac{\Delta I_N}{\|\Delta I_N\|_\infty}$

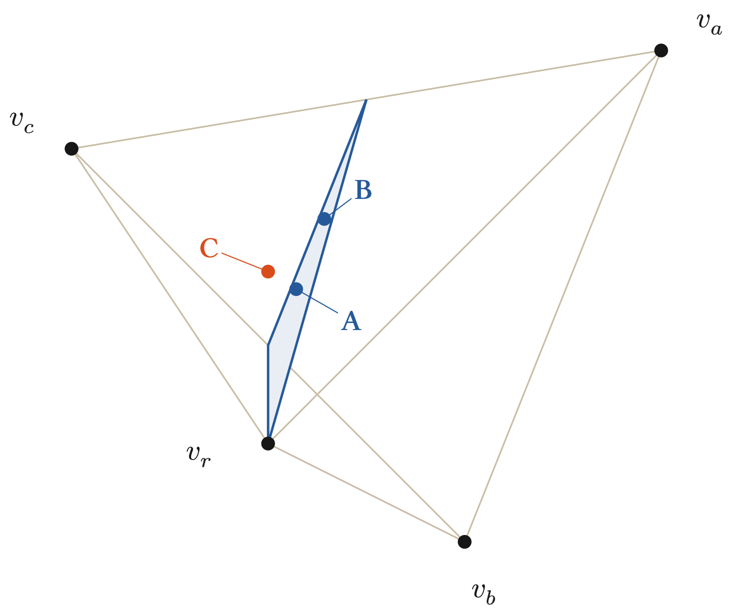
$$u_{3,k} = \frac{8u_{4,k} + 6u_{2,k}}{14}, \quad \Delta u_k = \frac{u_{4,k} + u_{2,k}}{2} - u_{3,k}$$

Proved-derived / exact coefficient witness. Here $c = 2, 3, 4$ selects the displayed output port and k counts diffusion steps. Linear diffusion preserves the weighted union at every step. Intensity is read after currents combine. The last column shows $42\Delta u_0$ and $\frac{\Delta I_N}{\|\Delta I_N\|_\infty}$; gray uses the same $I_\star$ as plate 6. The exact bound $\|\Delta\lambda\|_1 = \frac{1}{42}$ gives $|\Delta u_k| \le \frac{1}{42}$ and $|\Delta I_k| \le \frac{1}{21}$.

PLATE 08

Real strings as a tetrahedral current

Actual inscriptions supply the stimulus; the chart differentiates their input, output and reference ports.



$$v_a = (1,1,1), \quad v_b = (1,-1,-1)$$
$$v_c = (-1,1,-1), \quad v_r = (-1,-1,1)$$

Regular tetrahedron. $v_k \cdot v_k = 3$ and $v_k \cdot v_l = -1$ for $k \neq l$; every squared edge length is 8. The shaded section carries the addition constraint.

$$\delta(a,b,c) = a+b-c, \quad \hat{\delta}(p) = \frac{1+3p_x-p_y+p_z}{1-p_x-p_y+p_z} = \delta(q^{-1}(p))$$

$$\delta = 0 \quad \Leftrightarrow \quad \lambda_a + \lambda_b = \lambda_c \quad \Leftrightarrow \quad 1+3p_x-p_y+p_z = 0$$

Proved-derived / exact rational witness. A and B share the addition face but retain different geometric points. C has defect -1. The drawing uses camera $(x,y,z) \mapsto (x+\frac{z}{2}, y+\frac{z}{4})$; the complete rational 3D points remain in the table and source receipt.

Three source inscriptions, one explicit codec

	String	$p = q(a,b,c)$	m	δ
A	1+2=3	$\frac{\begin{pmatrix}-1\\1\\-3\end{pmatrix}}{7}$	7	0
B	2+1=3	$\frac{\begin{pmatrix}-1\\3\\-1\end{pmatrix}}{7}$	7	0
C	1+2=4	$\frac{\begin{pmatrix}-1\\1\\-2\end{pmatrix}}{4}$	8	−1

$$\tilde{h} = (a,b,c,1), \quad m = a+b+c+1, \quad \lambda = \frac{\tilde{h}}{m}$$
$$q(a,b,c) = \lambda_a v_a + \lambda_b v_b + \lambda_c v_c + \lambda_r v_r$$
$$\lambda_k = \frac{1+v_k \cdot p}{4}, \quad (a,b,c) = \frac{(\lambda_a, \lambda_b, \lambda_c)}{\lambda_r}$$

Definition / exact source chart. Canonical strings `a+b=c` enter at integer currents. The chart extends to $h = (a,b,c) \in \mathbb{Q}^3_{\geq 0}$. The reference weight 1 makes $\lambda_r > 0$, so the triple is recoverable. Parsing remains exterior.

PLATE 09

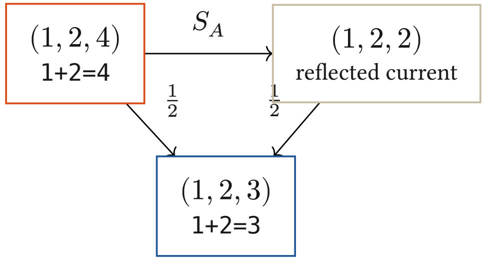
Reflection closes the same face in both charts

The integration law is explicit; the normalized geometric chart transports its weights.

Hold the two inputs; reflect the output current

$$S_A(a,b,c) = (a,b,2(a+b)-c), \quad S_A^2 = I$$

$$U_A = \frac{I+S_A}{2}, \quad U_A(a,b,c) = (a,b,a+b)$$



$$\delta(S_A h) = -\delta(h), \quad \delta(U_A h) = 0$$

Proved-derived. The arithmetic reflection reverses the signed defect. Its midpoint closes the relation while retaining a and b. This example integrates a current under a supplied addition law.

Face-conserving integration

$$\Delta_q(h) = q(U_A h) - U_B(q(h)) = 0$$

$$\hat{\delta}(U_B(q(h))) = \delta(U_A h) = 0$$

Same source, same declared comparison, same resulting face. The induced geometric update is now exhibited, rather than left as an unspecified condition.

The complete square now commutes

$$\begin{tikzcd}
(1,2,4) \arrow[r, "U_A"] \arrow[d, "q"] & (1,2,3) \arrow[d, "q"] \\
p_4 = \frac{(-1,1,-2)}{4} \arrow[r, "U_B"] & p_3 = \frac{(-1,1,-3)}{7}
\end{tikzcd}$$

$$p_2 = q(1,2,2) = \frac{(0,0,-1)}{3}$$

$$U_B(q(h)) = \frac{m(h)q(h) + m(S_A h)q(S_A h)}{m(h) + m(S_A h)}$$

$$U_B(p_4) = \frac{8p_4 + 6p_2}{14} = p_3$$

$$m_4 = 1+2+4+1 = 8, \quad m_2 = 1+2+2+1 = 6$$

Proved-derived. Normalization changes affine averaging. For h and $S_A h$ in the admitted chart, the weights follow from their source currents and reference port.

An untransported midpoint leaves a defect

$$p_{\text{naive}} = \frac{p_4 + p_2}{2}, \quad q^{-1}(p_{\text{naive}}) = \left(1, 2, \frac{20}{7}\right)$$

$$\hat{\delta}(p_{\text{naive}}) = 1 + 2 - \frac{20}{7} = \frac{1}{7} \neq 0$$

The Euclidean midpoint in this chart represents a different update. Retaining the chart weights restores the commuting square.

Exact rational witness. `arithmetic-example.py` generates the source receipt used by the figure and verifies reconstruction, reflection, closure, and the $\frac{1}{7}$ control. This is an exterior arithmetic construction, not a claim of learned HNN parsing or a general native training algorithm.